Hackathon Pitch Script: Value-Aware Churn Retention System

7-Minute Investor and Jury Narrative

Team Name Placeholder

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How to Use This Document

This is a speaker-ready pitch script designed for a **7-minute** presentation. It is not a copy of the methodology report. It is optimized for business impact first, then technical credibility.

Suggested timing plan:

- 0:00–0:50 Hook and problem framing
- 0:50–1:50 Customer journey story (2 personas)
- 1:50–3:00 Solution overview and operating loop
- 3:00–3:45 Model 1 and dataset (concise)
- 3:45–5:05 Model 2 and dataset (core innovation)
- 5:05–6:15 Proof of concept and results
- 6:15–7:00 Limitations, roadmap, and close

1 Hook and Problem Framing (0:00–0:50)

Opening line: Most banks already have churn scores. The real question is: *what action should we take now, for this customer, to maximize value?*

Today, retention is often done manually, by campaign rules or gut feeling:

- teams decide interventions ad hoc,
- expensive actions are overused on low-value customers,
- outcomes are weakly tracked and rarely fed back into decisioning.

Our thesis: churn prediction alone is not enough. We need a system that decides:

1. **who to target**, and
2. **which action** maximizes expected net profit.

So we optimize not for "fewest churns," but for **highest retention value after intervention cost**.

2 Customer Journey Story (0:50–1:50)

We tell the story through two customers flagged as high churn risk by Model 1.

Persona A: Low-value digital customer

- Uses one product, low balance, mostly app interactions, low expected annual value.
- Model 1 flags elevated risk based on recent behavior change.
- Model 2 evaluates all available actions and selects a **low-cost push notification with cashback offer**.

Why this is optimal: calling this customer would cost more than expected retained value. A lightweight nudge is economically rational.

Persona B: High-value relationship customer

- Multiple products, high balance, strong long-term contribution, sudden churn signals.
- Model 1 flags high risk and high value at risk.
- Model 2 selects **escalate to banker** (operationally represented by the high-touch action in our action set), with a **ticket and AI summary sent by email** to speed intervention quality.

Why this is optimal: high-touch action is expensive, but justified by expected value recovery.

Message to jury: same risk, different economics, different action. Based on the data, not on the gut feeling.

3 Solution Overview (1:50–3:00)

Our system is a 2-model operating loop:

1. **Activation by Model 1 (risk gate):**

$$r(x_i) = \Pr(Y_i = 1 \mid X = x_i), \qquad \text{activate if } r(x_i) > t.$$

2. **Action optimization by Model 2 (contextual bandit):** choose one action from $\mathcal{A}$ that maximizes expected net reward:

$$a_i = \arg\max_{a \in \mathcal{A}} \hat{\mu}_a(x_i), \qquad r_i = (1 - Y_i)\, v_i - \lambda c(a_i).$$

3. **Outcome after 90 days:** observe $Y_i \in \{0, 1\}$, where $Y_i = 1$ means full churn.
4. **Feedback loop:** update Model 2 parameters with observed reward:

$$\theta_{a_i} \leftarrow \theta_{a_i} + \eta\big(r_i - \theta_{a_i}^\top x_i\big) x_i.$$

Interpretation: first decide if intervention is worth considering, then choose the best action economically, then learn from delayed outcome.

Operational Loop (simple view)

$$\boxed{r(x_i) > t} \implies \boxed{a_i = \arg\max_{a \in \mathcal{A}} \hat{\mu}_a(x_i)} \implies \boxed{Y_i \text{ at } +90\text{d}} \implies \boxed{\theta_{a_i} \leftarrow \theta_{a_i} + \eta\big(r_i - \theta_{a_i}^\top x_i\big)x_i}$$

with

$$r_i = (1 - Y_i)\, v_i - \lambda c(a_i),$$

where v_i is customer value, $c(a_i)$ is action cost, and λ is cost sensitivity.

Figure 1: Risk-gated, value-aware retention decision loop used in the hackathon pipeline.

4 Model 1 + Dataset (3:00–3:45)

We keep this concise in the live pitch.

Dataset and setup:

- 9-month feature window, 90-day churn horizon.
- At-risk cohort: customers with active products at reference time.
- Size: $n = 7{,}805$, with 131 churn events (1.7%).

Modeling stack:

- LightGBM classifier,
- class imbalance handling with SMOTE (train folds only),
- Optuna tuning (150 trials, stratified CV),
- SHAP for explainability.

Performance (test):

- Best threshold $t = 0.88$,
- F1 = 0.5833 (precision 0.6364, recall 0.5385),
- improved from baseline F1 = 0.3860.

Most important signals (from engineered feature set): recency, product depth, transaction/balance dynamics, digital behavior, and complaint/contact intensity.

5 Model 2 + Dataset (3:45–5:05)

Activation flow

Model 2 runs only on activated events:

$$r(x_i) > t$$

This controls spend and avoids applying interventions to customers with marginal risk.

Action set and learning objective

For hackathon scope, we use:

$$\mathcal{A} = \{\text{no action},\ \text{push notification},\ \text{email},\ \text{call/escalation}\}.$$

Business interpretation:

- no action: preserve budget,
- push/email: scalable low-cost nudges,
- call/escalation: expensive but powerful high-touch intervention.

Objective (value-aware):

$$\pi^\star(x) \in \arg\max_{a \in \mathcal{A}} \mu_a(x), \qquad \mu_a(x) = \mathbb{E}\big[(1-Y)v - \lambda c(a) \mid X = x, A = a\big].$$

This is the key distinction: the policy learns *economic* uplift, not just churn reduction.

Outcome observation and update

At +90 days, we observe churn outcome and compute reward. For linear per-action models:

$$\hat{\mu}_a(x) = \theta_a^\top x,$$

only the chosen action is updated after observation:

$$\theta_{a_i} \leftarrow \theta_{a_i} + \eta\big(r_i - \theta_{a_i}^\top x_i\big)x_i.$$

Intuition for non-technical audience: every decision becomes new evidence; over time, the model allocates costly actions only where they pay off.

6 Proof of Concept Results (5:05–6:15)

This section summarizes `05_contextual_bandit.ipynb`.

Important context: this is a controlled offline experiment with known ground-truth expected rewards per action, enabling exact policy comparison.

Learning signals observed:

- Cumulative revenue saved vs no-action baseline in simulation: **€16,588,031**.
- Early months (1–6): average improvement vs no-action **+€132.8/customer**, call fraction **0.0%**.
- Late months (last 6): average improvement **+€290.8/customer**, call fraction **20.0%**.

Holdout (last 6 months):

- 16,386 rows (2025-07 to 2025-12),
- learned policy expected reward: **€3,940/customer**.

Action mix of learned policy (holdout):

- no action: 13.3%
- push notification: 27.6%
- email: 34.0%
- call/escalation: 25.1%

Net revenue saved per customer vs do-nothing:

- Learned policy: **+€ 292**
- Always email: +€ 117
- Always push: +€ 52
- Uniform random: -€ 5
- Always call: -€ 188

Interpretation: the model avoids the expensive "always call" trap and selectively deploys high-touch actions where value justifies cost.

Further diagnostics from notebook:

- Call fraction months 1–6: 5.0%; months 19–24: 25.0%.
- Call targeting by value decile: D1 = 0.0%, D10 = 25.1%.
- Call targeting by risk decile: R1 = 0.0%, R10 = 64.9%.

Projected impact (scenario in notebook):

- holdout-based annual savings: **€ 4,218,085**,
- steady-state annual savings estimate: **€ 4,388,748**,
- cumulative over 24 months: **€ 7,305,936**.

7 Future Steps and Limitations (6:15–7:00)

Current limitations (transparent):

- one-step bandit: one action per decision event,
- delayed reward and potential confounding from historical policies,
- dependence on logging quality and propensity data,
- risk-gating may hide effects outside activated population.

Roadmap:

1. Move from **single action** to **multi-step retention journeys** (action sequences and workflows).
2. Add richer causal/off-policy evaluation and stronger uncertainty calibration.
3. Upgrade action space with campaign-specific variants and channel orchestration.
4. Integrate banker copilot layer for faster, consistent high-touch execution.
5. Introduce policy guardrails (budget caps, fairness, compliance constraints) in optimization.

Closing Line

We built a system that does not stop at predicting churn. It decides the right action, for the right customer, at the right cost, and learns from outcomes. That is the path from analytics to measurable retention profit.